SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 12 / 25 completed

Sum Diagonal

SUM DIAGONAL

Write a C program to find the sum of the diagonal elements of a matrix.

A matrix is a two-dimensional array of numbers arranged in rows and columns. The diagonal elements of a matrix are the elements that lie on the main diagonal, which starts from the top-left corner and goes to the bottom-right corner of the matrix.

Sample Input:
Let's consider a 3x3 matrix as an example.
Enter the number of rows (max 100): 3
Enter the number of columns: 3
Enter the elements of the matrix:
1 2 3
4 5 6
7 8 9

Sample Output:
The matrix is:
1 2 3

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main() {
    int r, c, i, j, sum = 0;
    int a[100][100];

    printf("Enter the number of rows (max 100): ");
    scanf("%d", &r);

    printf("Enter the number of columns: ");
    scanf("%d", &c);

    printf("Enter the elements of the matrix:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nThe matrix is:\n");
    for (i = 0; i < r; i++) {
```

OUTPUT

```
Enter the number of rows (max 100): Enter the number of
columns: Enter the elements of the matrix:

The matrix is:
1 2 3
4 5 6
7 8 9
The sum of the diagonal elements is: 15
```

INPUT

```
3
3
1 2 3
4 5 6
```

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